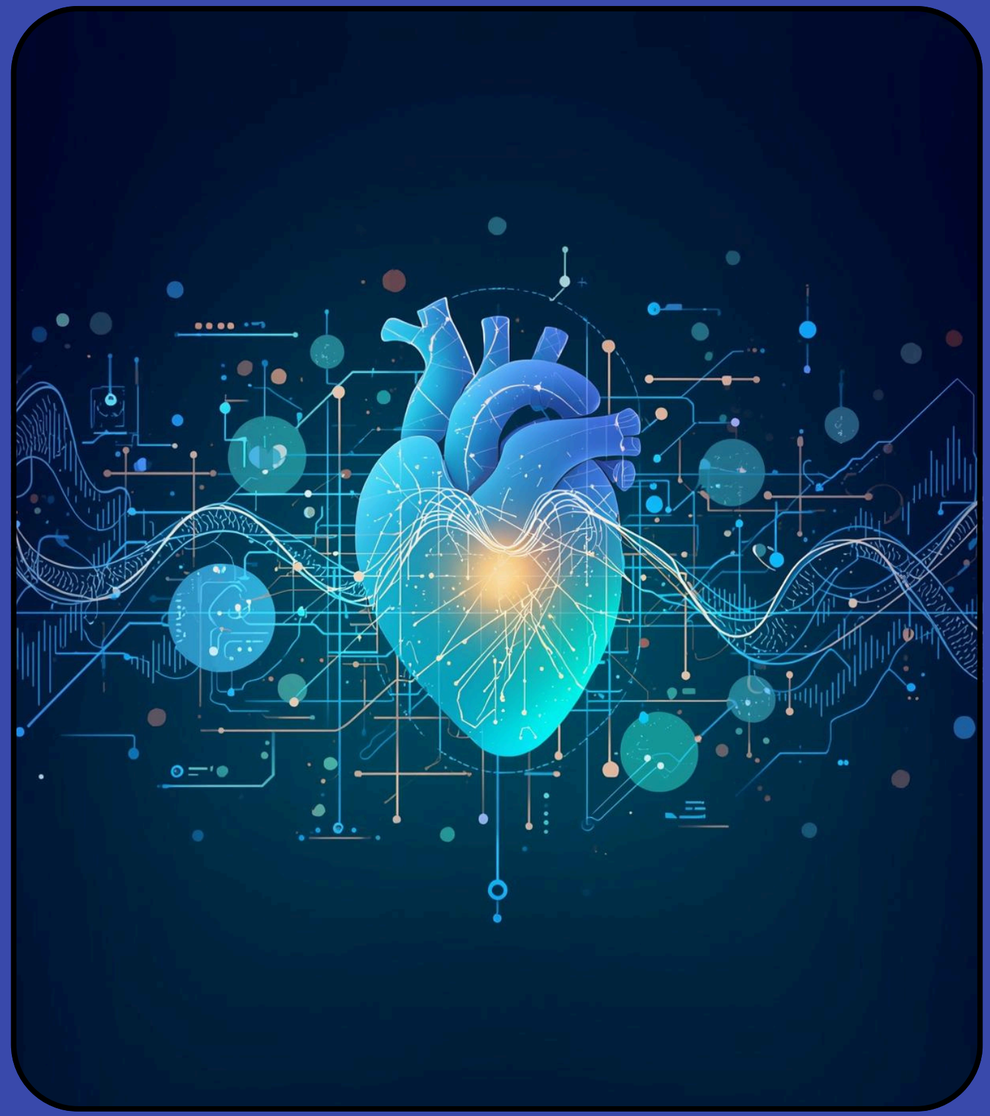
PREDICTING HEART HEALTH

# Heart Disease Prediction




Academic Group Project

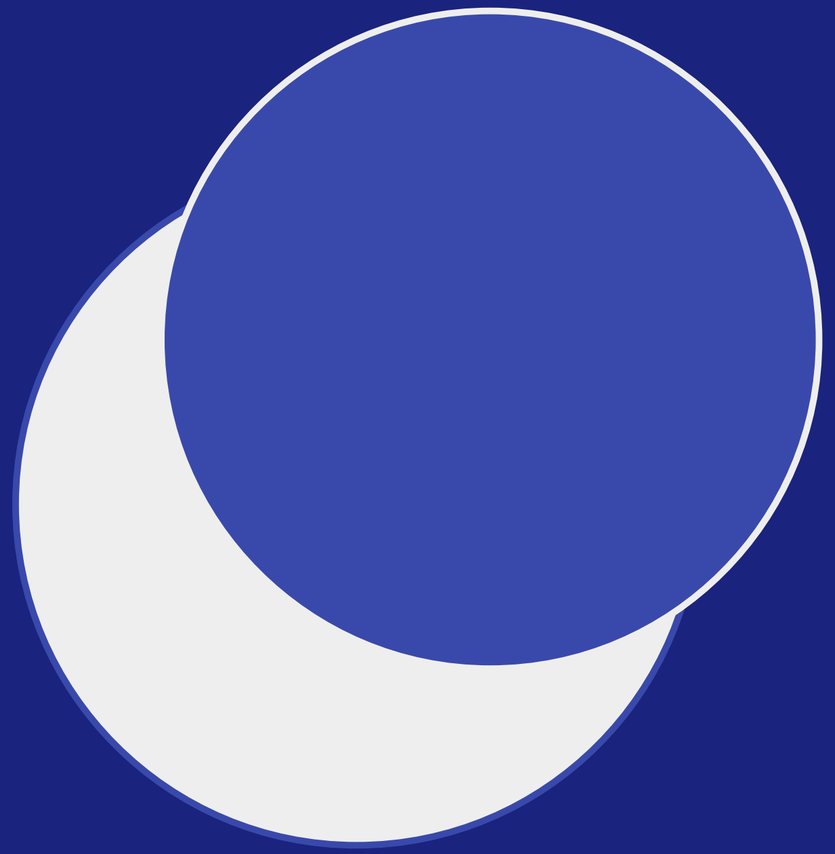
University of Jeddah

# Project Overview

- Heart disease is a major global health concern requiring early prediction.
- This project applies data mining techniques to predict heart disease risk.

# Data Understanding & Preparation

- The dataset includes clinical and demographic heart health attributes.
- Initial data inspection identified issues requiring preprocessing.

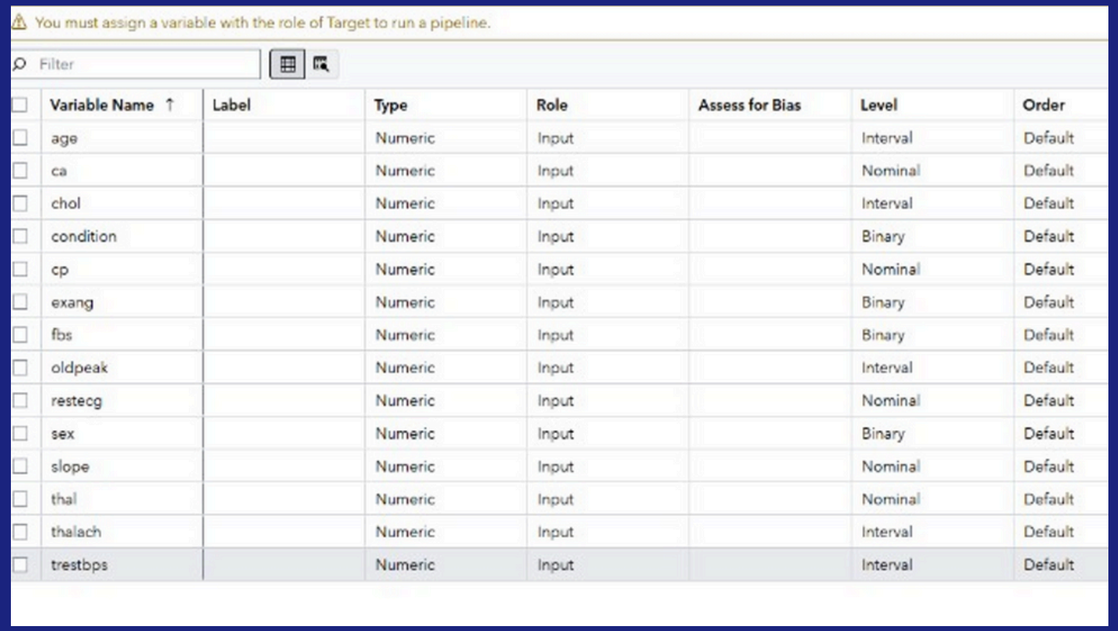


# Data Collection and Cleaning



# Data Collection and Cleanin

## Before cleaning

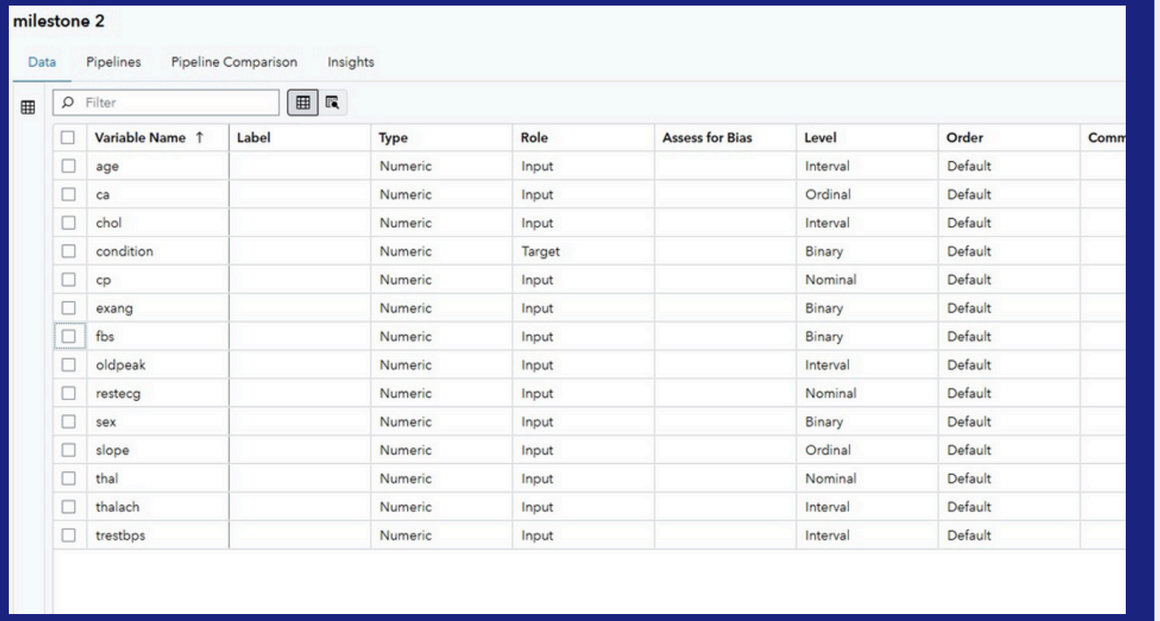


⚠ You must assign a variable with the role of Target to run a pipeline.

Filter

| Variable Name ↑ | Label | Type | Role | Assess for Bias | Level | Order |
|---|---|---|---|---|---|---|
| age | | Numeric | Input | | Interval | Default |
| ca | | Numeric | Input | | Nominal | Default |
| chol | | Numeric | Input | | Interval | Default |
| condition | | Numeric | Input | | Binary | Default |
| cp | | Numeric | Input | | Nominal | Default |
| exang | | Numeric | Input | | Binary | Default |
| fbs | | Numeric | Input | | Binary | Default |
| oldpeak | | Numeric | Input | | Interval | Default |
| restecg | | Numeric | Input | | Nominal | Default |
| sex | | Numeric | Input | | Binary | Default |
| slope | | Numeric | Input | | Nominal | Default |
| thal | | Numeric | Input | | Nominal | Default |
| thalach | | Numeric | Input | | Interval | Default |
| trestbps | | Numeric | Input | | Interval | Default |

## After cleaning



milestone 2

Data Pipelines Pipeline Comparison Insights

Filter

| Variable Name ↑ | Label | Type | Role | Assess for Bias | Level | Order | Comm |
|---|---|---|---|---|---|---|---|
| age | | Numeric | Input | | Interval | Default | |
| ca | | Numeric | Input | | Ordinal | Default | |
| chol | | Numeric | Input | | Interval | Default | |
| condition | | Numeric | Target | | Binary | Default | |
| cp | | Numeric | Input | | Nominal | Default | |
| exang | | Numeric | Input | | Binary | Default | |
| fbs | | Numeric | Input | | Binary | Default | |
| oldpeak | | Numeric | Input | | Interval | Default | |
| restecg | | Numeric | Input | | Nominal | Default | |
| sex | | Numeric | Input | | Binary | Default | |
| slope | | Numeric | Input | | Ordinal | Default | |
| thal | | Numeric | Input | | Nominal | Default | |
| thalach | | Numeric | Input | | Interval | Default | |
| trestbps | | Numeric | Input | | Interval | Default | |

# Data Collection and Cleanin

## Statistics before find outliers

| Missing | Minimum | Maximum | Mean |
|---|---|---|---|
| 0.0000 | 29.0000 | 77.0000 | 54.5421 |
| 0.0000 | 0.0000 | 3.0000 | 0.6768 |
| 0.0000 | 126.0000 | 564.0000 | 247.3502 |
| 0.0000 | 0.0000 | 1.0000 | 0.4613 |
| 0.0000 | 0.0000 | 3.0000 | 2.1582 |
| 0.0000 | 0.0000 | 1.0000 | 0.3266 |
| 0.0000 | 0.0000 | 1.0000 | 0.1448 |
| 0.0000 | 0.0000 | 6.2000 | 1.0556 |
| 0.0000 | 0.0000 | 2.0000 | 0.9966 |
| 0.0000 | 0.0000 | 1.0000 | 0.6768 |
| 0.0000 | 0.0000 | 2.0000 | 0.6027 |
| 0.0000 | 0.0000 | 2.0000 | 0.8350 |
| 0.0000 | 71.0000 | 202.0000 | 149.5993 |
| 0.0000 | 94.0000 | 200.0000 | 131.6936 |

## Statistics after find outliers

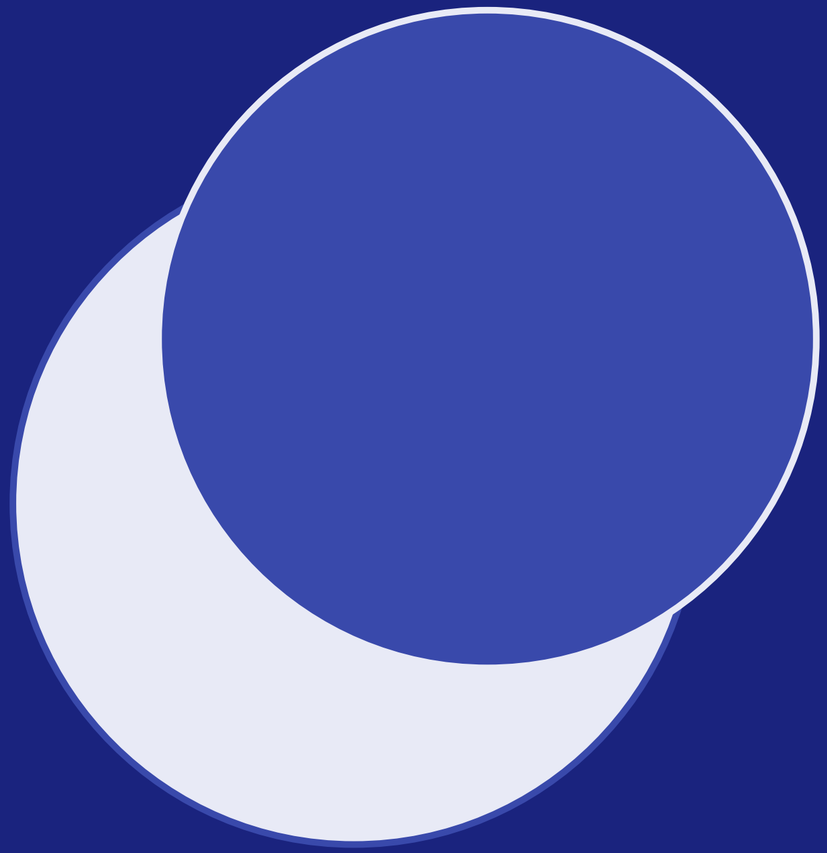
The SAS System

The MEANS Procedure

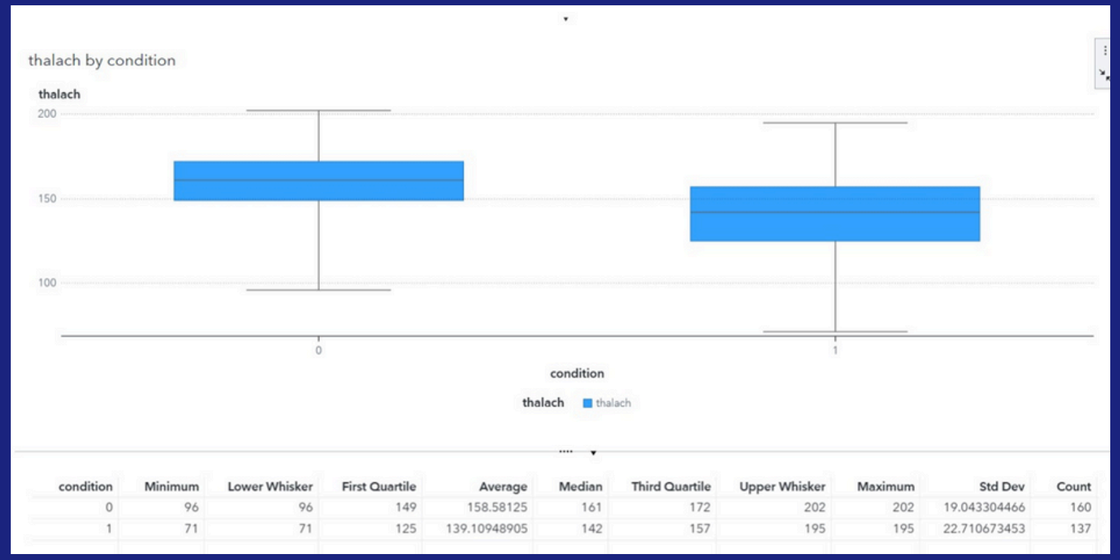
| Variable | Label | N | Mean | Std Dev | Minimum | Maximum |
|---|---|---|---|---|---|---|
| condition | | 286 | 0.4510490 | 0.4984702 | 0 | 1.0000000 |
| age | | 286 | 54.3811189 | 9.1370574 | 29.0000000 | 77.0000000 |
| sex | | 286 | 0.6853147 | 0.4652044 | 0 | 1.0000000 |
| cp | | 286 | 2.1503497 | 0.9634277 | 0 | 3.0000000 |
| trestbps | | 286 | 130.9440559 | 16.8424177 | 94.0000000 | 180.0000000 |
| chol | | 286 | 244.1538462 | 45.8012193 | 126.0000000 | 394.0000000 |
| fbs | | 286 | 0.1433566 | 0.3510504 | 0 | 1.0000000 |
| restecg | | 286 | 0.9755245 | 0.9961834 | 0 | 2.0000000 |
| thalach | | 286 | 149.7167832 | 23.0087748 | 71.0000000 | 202.0000000 |
| exang | | 286 | 0.3251748 | 0.4692612 | 0 | 1.0000000 |
| oldpeak | | 286 | 0.9671329 | 1.0305356 | 0 | 4.0000000 |
| slope | | 286 | 0.5769231 | 0.5976788 | 0 | 2.0000000 |
| ca | | 286 | 0.6398601 | 0.9059367 | 0 | 3.0000000 |
| thal | | 286 | 0.8006993 | 0.9507889 | 0 | 2.0000000 |
| _dmIndex_ | | 286 | 30748254.59 | 17990238.31 | 1000001.00 | 64000004.00 |
| _PartInd_ | Partition Indicator | 286 | 0.8006993 | 0.6030329 | 0 | 2.0000000 |

# Exploratory Data Analysis

- Visual analysis revealed clear differences between healthy and affected patients.
- Key variables showed strong influence on heart disease presence.

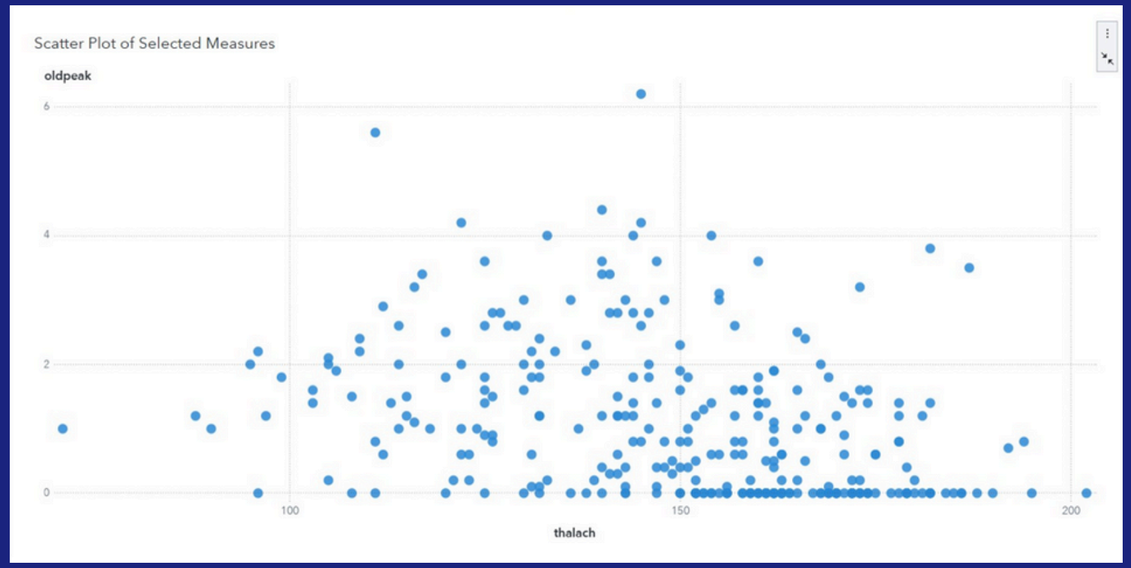


# Exploratory Data Analysis




| condition | Minimum | Lower Whisker | First Quartile | Average | Median | Third Quartile | Upper Whisker | Maximum | Std Dev | Count |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 96 | 96 | 149 | 158.58125 | 161 | 172 | 202 | 202 | 19.043304466 | 160 |
| 1 | 71 | 71 | 125 | 139.10948905 | 142 | 157 | 195 | 195 | 22.710673453 | 137 |

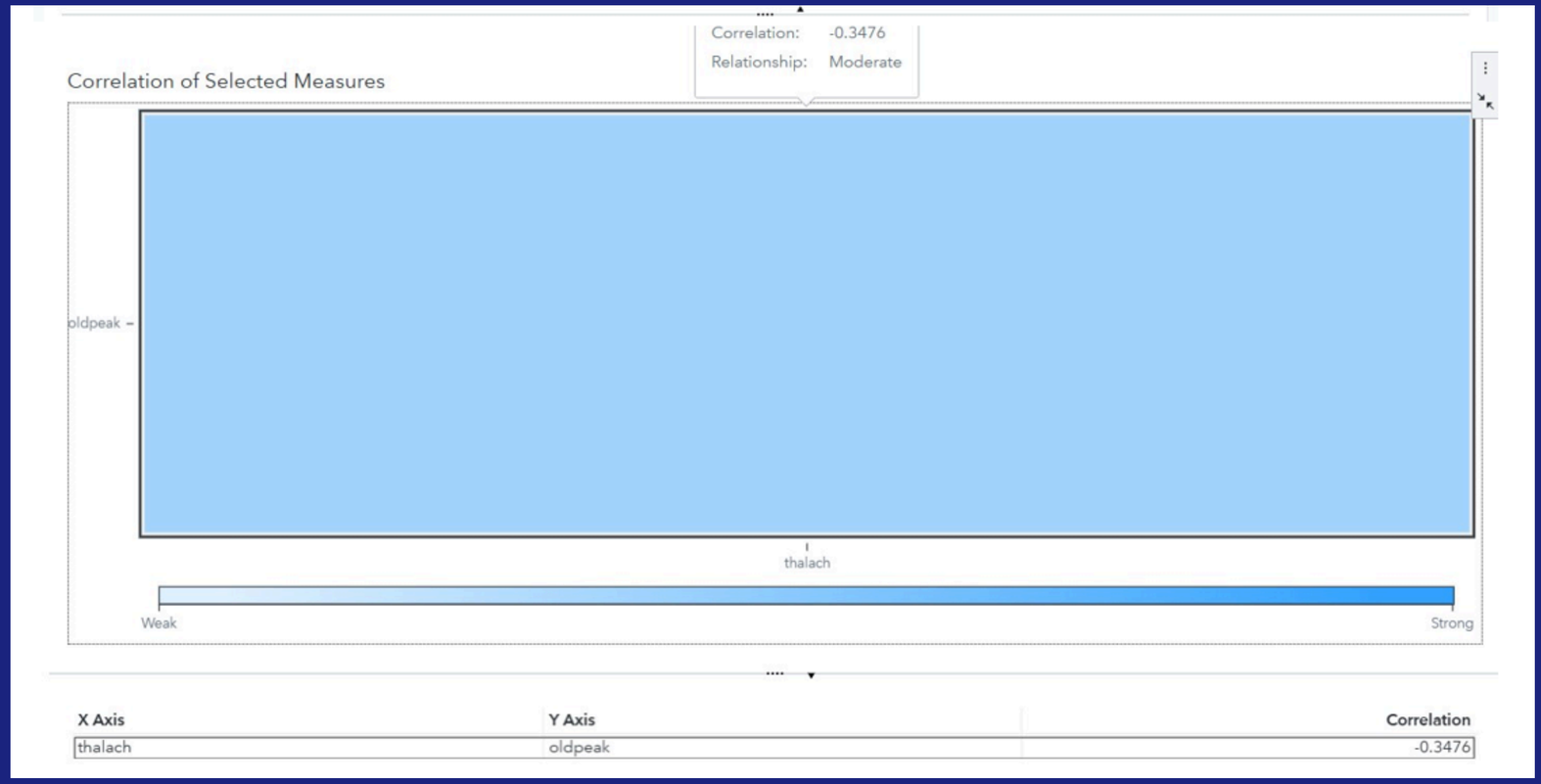
Box plot (thalach by condition)




Scatter plot (oldpeak vs thalach)

# Exploratory Data Analysis

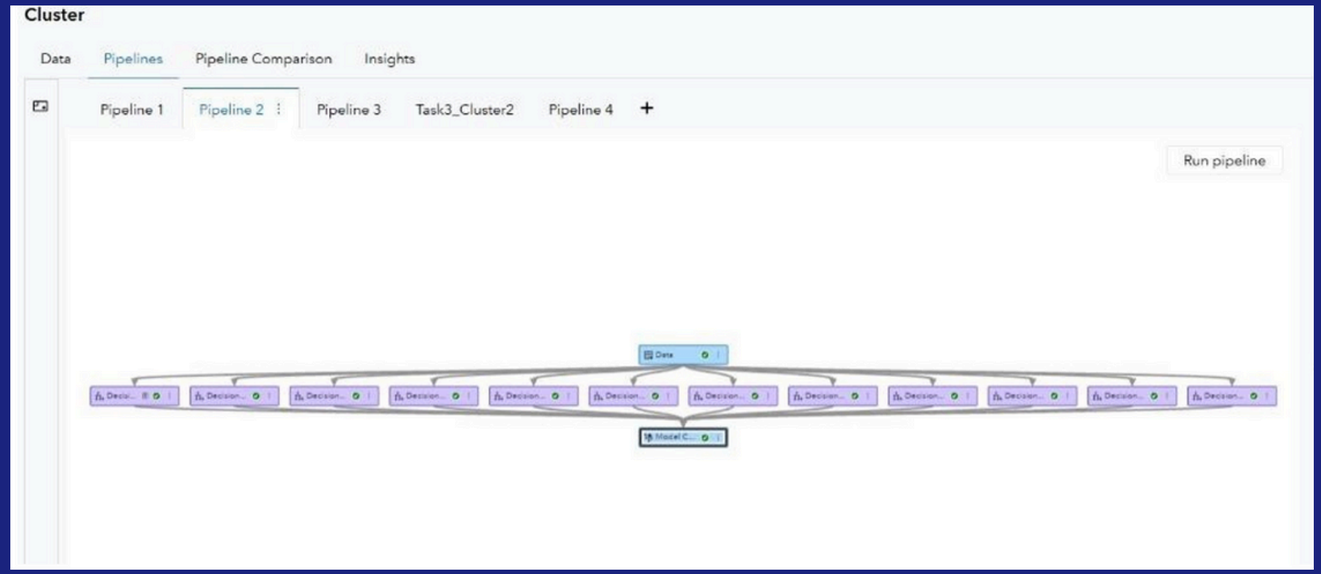
## Correlation Matrix




| X Axis | Y Axis | Correlation |
|---|---|---|
| thalach | oldpeak | -0.3476 |

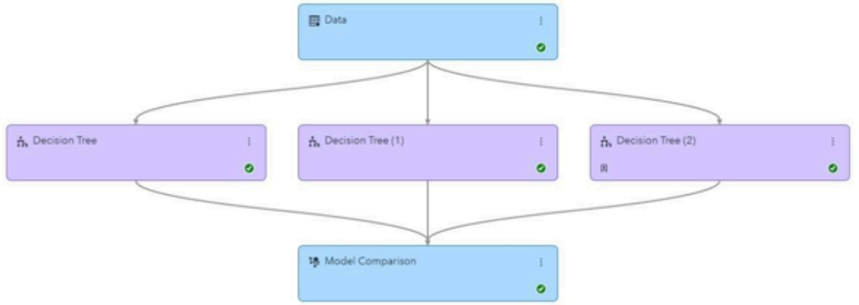
# Decision Tree Modeling

- Multiple Decision Tree models were built using different splitting criteria.
- Models were evaluated using accuracy, lift, and stability measures.



# Best Model Selection

- Decision Tree (2) achieved the best balance between performance and simplicity.
- The selected model showed consistent and interpretable results.

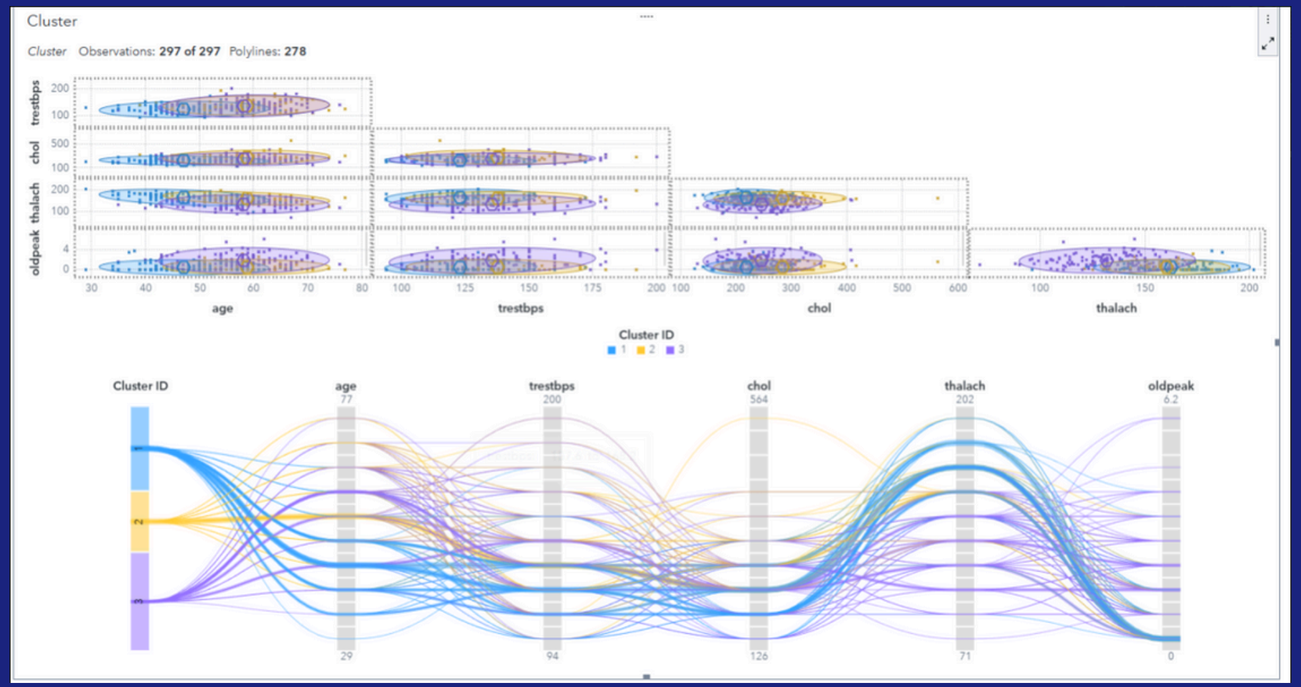



**Model Comparison**

| Cha... | Name |
|---|---|
| ★ | Decision Tree (2) |
| | Decision Tree |
| | Decision Tree (1) |

# K-Means Clustering

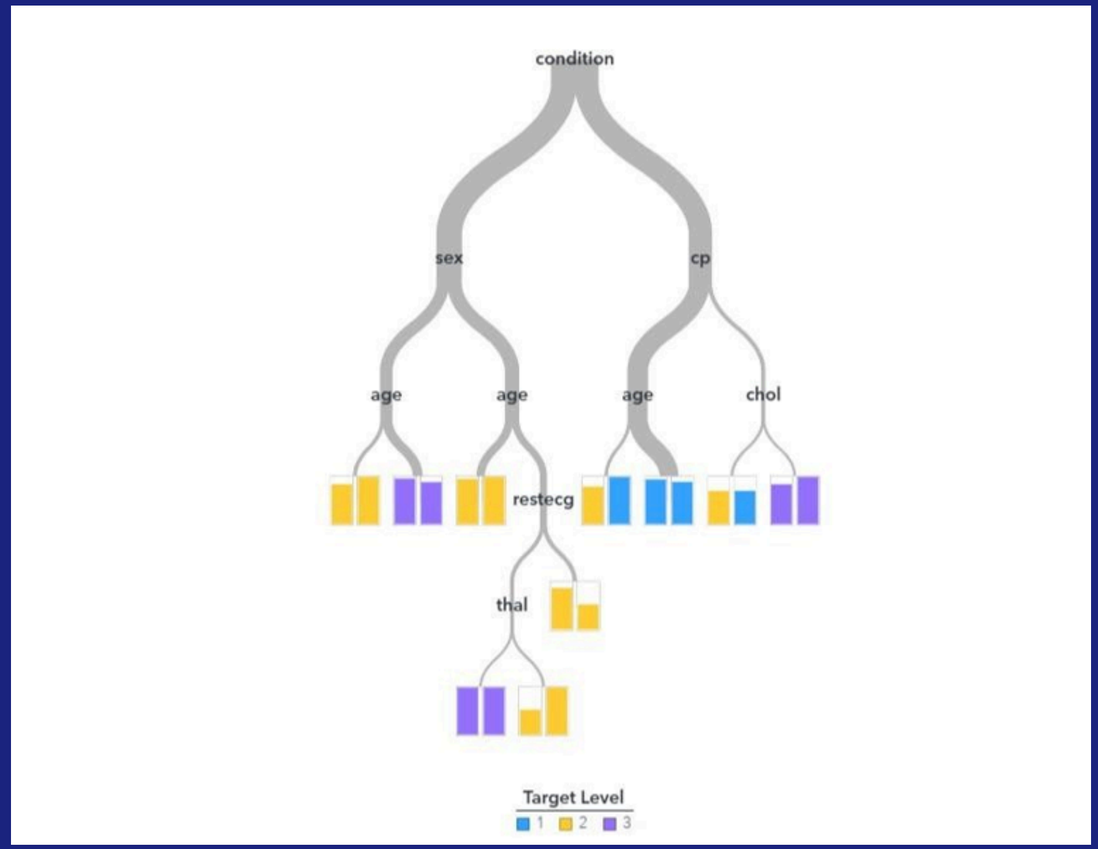
- K-means clustering was applied to segment patients into groups.
- The km_3 segmentation provided the most meaningful patient clusters.




| Segmentation | Clusters | Cluster Sizes | Smallest Cluster | Outliers |
|---|---|---|---|---|
| km_2 | 2 | (119, 156) | 119 | no |
| km_3 | 3 | (82, 113, 80) | 80 | no |
| km_4 | 4 | (84, 77, 57, 57) | 57 | no |
| km_5 | 5 | (41,49,57,56, 72) | 41 | no |

# Cluster-Based Decision Trees

- Separate Decision Trees were built for the largest cluster.
- Cluster-specific models improved prediction accuracy.



# Conclusion

- Data preprocessing played a key role in improving model performance.
- Decision Trees provided clear and interpretable predictions.
- K-means clustering enhanced understanding of patient subgroups.
- Combining clustering with classification improved overall results.

# Thank You